

Visual Partitions versus Per-Element Marking on Solid T-Beams

Grok Build ¹

¹ Computational mechanics campaign on CalculiX C3D8 T-beams.

Abstract

On a frozen-topology solid T-beam, every classical adaptive cycle pays for one global assembly and solve. Visual partitioning names a few structural regions, writes a size on each, and rebuilds a conforming mesh three times. This paper compares those two discretisation philosophies on the same object: a three-dimensional C3D8 cantilever in CalculiX 2.23. The classical side is per-element Zienkiewicz–Zhu / residual Dörfler marking ($\theta = 1/2$) with hanging 8-splits, and per-element Houston–Wihler local hp prediction ($\theta = 1/3$), which is not a partition. The visual side is a log-linear size field through named regions (clamp face, flange–web re-entrant corner, tip flange, plus a distinctive feature on each taper), one residual revision, and a one-generation two-coordinate particle swarm. Error is measured against a uniform $h = 20$ mm C3D8 reference, not against Euler–Bernoulli theory. On prismatic G01 under linear static load, three partition solves give $e_u = 3.58\%$, 2.90% , 2.91% at $N = 5184$, 6048 , 6048 equations. Per-element ZZ–Dörfler at the same three global solves remains at 16.1% , 15.5% , 13.7% . The Dörfler curve is continued to eight points until an equation cap; it is not a three-point method and it is not a partition. A six-zone spanwise template is not a method of this paper.

Keywords: adaptive mesh refinement; T-beam; visual partition; size field; Dörfler marking; Houston–Wihler; CalculiX

1. Introduction

A solid T-beam is not a model elliptic equation on a uniform grid. The clamp face, the flange–web re-entrant corner, and the free-end load introduction each ask for local resolution. Uniform hexahedra spend most of their degrees of freedom in the smooth mid-span. Adaptive mesh refinement answers that request by a loop: solve, estimate, mark, refine [1–3]. Dörfler’s object is the element K [2]. Houston and Wihler’s object is the face-neighbour patch of each hex κ [4]. Neither paper partitions the domain into named structural blocks.

Engineers often do the opposite. They look at the structure, name a few parts, and write a size on each part. The mesh is then generated from a size field. That is a partition. A short loop—vision sizes, one residual revision, two-coordinate calibration—uses three global solves. The scientific question is where that short loop sits relative to a per-element marking loop, read at matched equation counts *and* at matched numbers of global solves.

This paper reports one CalculiX campaign on that question. It does not treat Dörfler as a partition, it does not treat Houston–Wihler as a partition, and it does not revive a six-zone spanwise template that exists in the repository as a 2026-08-15 coding artefact. The contribution is a comparison of two discretisation philosophies on one solid T-beam family, not a new graph network.

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2. Related Work

2.1. Per-element h -adaptivity

Dörfler proved that bulk marking by a collective fraction θ of $\sum \eta_K^2$, followed by refinement and re-solve, converges for Poisson [2]. The marked object is the element. Morin, Nochetto and Siebert identified data oscillation [5]; Cascón et al. obtained quasi-optimal rates [6]. None of those theorems define the tip displacement after two cycles as the truth. This paper takes the 1996 element object, $\theta = 1/2$, and hanging 8-split as the classical h protocol. Simple nodal-averaging Zienkiewicz–Zhu recovery [1] is the working indicator (not superconvergent patch recovery). Residual indicators on C3D8 one-point stresses are run as a separate loop. Amor-Martín and Garcia-Castillo discuss conforming steps that avoid hanging nodes in prismatic electromagnetics [7]; that literature is not a licence to rewrite Dörfler’s object as a spanwise station cut.

2.2. Local energy prediction is not a partition

Houston and Wihler pose a pair of local Dirichlet problems on $T_K^N = \{K\} \cup \text{face neighbours}$, compare predicted drops in potential $E_D = \frac{1}{2}a(u, u) - \ell(u)$, and mark by the maximal rule (1) with $\theta = 1/3$ [4]. Algorithm 1 line 14 returns to the global solve. Example 1 of that paper reports an hp mesh after nine adaptive refinements. The patch travels with K ; it is not a partition of Ω . Babuška and Rheinboldt used vertex-star Dirichlet problems [8]. Bammer, Schröder and Wihler later wrote an abstract loop [9]. This paper keeps Houston–Wihler marking at (4.12) and does not mix Dörfler into that outer loop. On a three-dimensional face-neighbour patch, quadratic serendipity hexes have every node on $\partial D(K)$, so the p -problem is empty. The p -branch here is an in-process 27-node $Q2$ assembly; CalculiX 2.23 is never sent TYPE=C3D27.

$$\Delta \tilde{E}_{K,\max} > \theta \max_{K'} \Delta \tilde{E}_{K',\max}, \quad \theta = \frac{1}{3}. \quad (1)$$

2.3. Size fields and learning

MeshingNet, LAMG and AMBER predict a size field and often retain only a global scalar to meet a budget. Loseille–Alauzet continuous mesh theory treats an embedded family as a global scaling of a fixed anisotropic shape. The two-coordinate particle swarm in this paper is that remaining freedom after a regional step, not the partition itself. Yang, Foucart, and ASMR write *marking* as a Markov decision process with the element as the agent [10,11]. Those works are learned per-element markers, not partitions. They are discussed, not retrained, here.

3. Materials and Methods

3.1. Geometry, material, load cases

Units are mm, N, MPa, t, s. The solver is CalculiX 2.23 with SPOOLES, entry /workspace/agent-ic-work/ccx, binary SHA-256 31be21fc2f0902bd9a05acc2651dbac6dc2a2573d Global elements are C3D8. Through-T counts are frozen at web 2×5 , outrigger 4, flange thickness 2, passed explicitly so automatic through-counts cannot add layers.

Prismatic G01 is the extrusion of flange 240×20 and web 20×140 over $L = 2000$ mm, volume 1.52×10^7 mm³, from tbeam_solid_hex.py. T01–T10 are the ten families in tbeam_taper.py. T01 is a linear haunch (web height $200 \rightarrow 80$ mm with a level top at $z = 220$), not an alias of G01. The coarse identity T0 is uniform_x_stations(18): 18 spanwise cells, 540 hexes, 912 nodes, 2592 unconstrained equations, verified on all eleven beams.

Steel is illustrative: $E = 2.1 \times 10^5$ MPa, $\nu = 0.3$, $\rho = 7.85 \times 10^{-9}$ t mm⁻³, $f_y = 235$ MPa. Five load cases: S-LIN (−5000 N on the tip-top set), S-ISO and S-KIN (plastic resultant from the root-section fibre formula with factor 1.35), D-LIN (−8000 N) and D-NLG (−30000 N) with the same pulse. S-KIN's second step is $-0.5F$, not a full reversal. Houston–Wihler is run only on S-LIN.

The S-LIN error is

$$e_u = \frac{|u_z - u_z^{\text{ref}}|}{|u_z^{\text{ref}}|}, \quad u_z = \text{mean}\{U_3(n) : n \in \text{TIP_TOP}\}, \quad (2)$$

with u_z^{ref} from a uniform $h = 20$ mm C3D8 mesh on the same section law (100 spanwise cells). Euler–Bernoulli tip displacement is tabulated as an order-of-magnitude column and is not an error axis.

3.2. Visual partition

G01 S-LIN uses three named regions: clamp face ($x_i = 40$ mm, $h = 42$), flange–web junction (120, 50), tip–flange load (1960, 64). Other load cases add a fourth structural region. Each taper adds one distinctive feature. Region count follows the view; it is not six. The size field is log-linear in x between ordered anchors (x_i, h_i) , constant outside the first and last anchor. Spanwise stations are generated by `tbeam_taper.build_x_stations` applied to that field. Vision box mid-points are not used as station generators.

After the delegated solve, residual $\eta_{K,R}^2$ is summed in spanwise Voronoi cells of the anchors. One call of `algorithm_from_mapping.run` with the default `AlgorithmConfig` (`pso_generations = 1`, seed 17) produces a mother-particle size vector and a selected-particle vector. Each is remeshed from scratch and solved. The three equation counts are required to be distinct; collisions are recorded and not plotted as fake iterations. Dörfler is not cancelled when the third partition point collides.

3.3. Per-element Dörfler

Indicators are $\eta_{K,ZZ}^2$ from simple recovery and $\eta_{K,R}^2$ from face jumps, computed per hex id. Marking is `doerfler_mark` with $\theta = 1/2$. Refinement is `refine_marked_hexes` with 1-irregular closure and `CalculiX *EQUATION`. Spanwise station bisection is not used. The loop stops at the first of: empty mark set; $N \geq N_{\text{cap}}$; twelve refinements. $N_{\text{cap}} = \min(4N_{\text{PSO}}, N_{\text{ref}})$ when a distinct third partition point exists, otherwise N_{ref} . Relative drops in e_u are diagnostic, not a stop. G01 ZZ is required to produce at least eight points unless the mark set empties or N_{ref} is hit.

3.4. Houston–Wihler

Local h is an 8-split of T_K^N . Local p is in-process Q2. After the first hanging 8-split, T_K^N uses matching-face adjacency plus the undirected hanging-face neighbourhood from the coarser-neighbour geometry test. Hanging slave nodes are removed from the local free set. Global mixed-order meshes are not hung: every marked hex, including p -winners, is 8-split globally, and $n_{p\text{-realized-as-}h} = |p\text{-marked}|$ is recorded. R8 checks four conformal seeds (clamp web, re-entrant, mid-span, tip flange) plus one hanging-interface seed.

4. Results

All numbers below come from `artifacts/amr_plan_run/` of this run. Old manuscript percentages are not reused.

4.1. References and T0

On G01 S-LIN the $h = 20$ mm reference has $N = 14400$ equations and $u_z^{\text{ref}} = -3.9674$ mm. T0 has $N = 2592$, $u_z = -3.3288$ mm, $e_u = 16.10\%$. The same T0 counts (540/912/2592) hold on T01–T10. A G01 $h = 28$ mm C3D8 card has $N = 10368$. A C3D20 card on the T0 layout has $N = 9573$ and completed. Reaction force on the clamp closes to the applied load at relative level 10^{-3} or better on the linear static cards.

T01 is not G01: its reference tip displacement is -2.7185 mm against G01's -3.9674 mm.

4.2. Visual partition, S-LIN

Table 1 lists the three partition solves. On G01 the delegated, revision and PSO meshes have $N = 5184, 6048, 6048$. The last two collide; that collision is recorded and the two points are not drawn as a fake third iterate. e_u is 3.58%, 2.90%, 2.91%. Six of the ten tapers produce three distinct N .

Table 1. S-LIN visual-partition short loop. e_u relative to the $h = 20$ mm C3D8 reference. Distinct N is a gate, not a smoothness claim.

Beam	N_{del}	N_{rev}	N_{pso}	distinct	e_u del/rev/ps0	(s, κ)
G01	5184	6048	6048	no	3.58 / 2.90 / 2.91	(0.00, 0.04)
T01	5184	5328	5328	no	3.53 / 3.63 / 3.69	(0.00, 0.04)
T02	5184	5760	5904	yes	5.11 / 3.98 / 3.66	(−0.039, 0)
T03	5040	5328	5328	no	6.96 / 5.38 / 5.38	(0.00, −0.04)
T08	5184	5328	5328	no	3.48 / 3.33 / 3.36	(0.00, 0.04)
T09	5184	5904	5760	yes	8.33 / 5.87 / 6.01	(0.039, 0)

4.3. Per-element hanging Dörfler

G01 ZZ–Dörfler ran eight cycles (T0 through cycle 7) and stopped on the equation cap (Table 2). Marked-hex counts are 33, 132, 130, 165, 212, 297, 395, 1172. e_u falls from 16.1% to 6.11%. It does not jump to a few percent in two cycles. T01 and T08 were run to the same stop (eight and six points). T02–T07, T09, T10 are T0 plus two cycles (existence), labelled as such in Figure 3. G01 residual Dörfler (hanging-face jumps not area-weighted; the stop was not changed) has six points, last $e_u = 7.76\%$ at $N = 20265$. A single uniform 8-split of every T0 hex produces 4320 hexes, $N = 16740$, $e_u = 4.10\%$.

Table 2. G01 S-LIN hanging ZZ–Dörfler. Encryption atom: 8-split of marked hexes plus 1-irregular *EQUATION. Not station bisection.

cycle	N	n_{hex}	n_{marked}	e_u
0	2592	540	33	0.1610
1	3216	771	132	0.1553
2	4371	1184	130	0.1373
3	5484	1618	165	0.1232
4	6999	2227	212	0.1143
5	8970	3018	297	0.1028
6	12174	4250	395	0.0833
7	16911	6077	1172	0.0611

4.4. Matched solves and matched N

Figure 1 is the comparison the question asks for. The partition abscissa is 1, 2, 3. Dörfler is counted from cycle 0. At three global solves, G01 partition e_u is 2.9%; ZZ–Dörfler is 13.7%. Figure 2 uses $n_{\text{equations}}$. At $N \approx 5500$ –6000, partition $e_u \approx 2.9\%$ while ZZ at $N = 5484$ is 12.3%. The two meshes do not share an atom: the caption of each figure states

that the partition mesh is a conforming size field and the Dörfler mesh is hanging 8-split. A uniform conforming ladder on G01 gives $e_u = 16.1\%$, 5.48% , 2.21% at $N = 2592, 4608, 6912$.

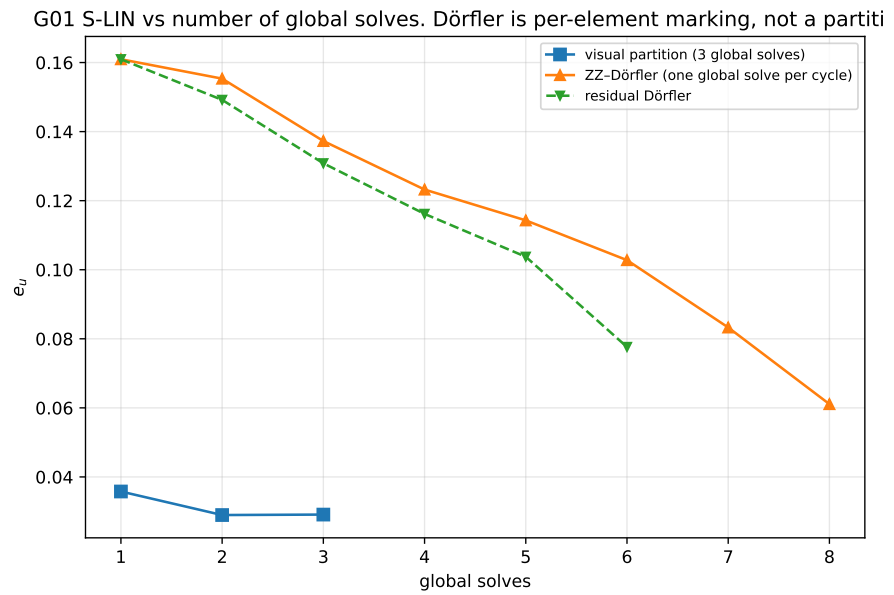


Figure 1. G01 S-LIN, e_u against the number of global solves. Blue squares: visual partition. Orange triangles: per-element ZZ-Dörfler (hanging 8-split). Dörfler is not a partition.

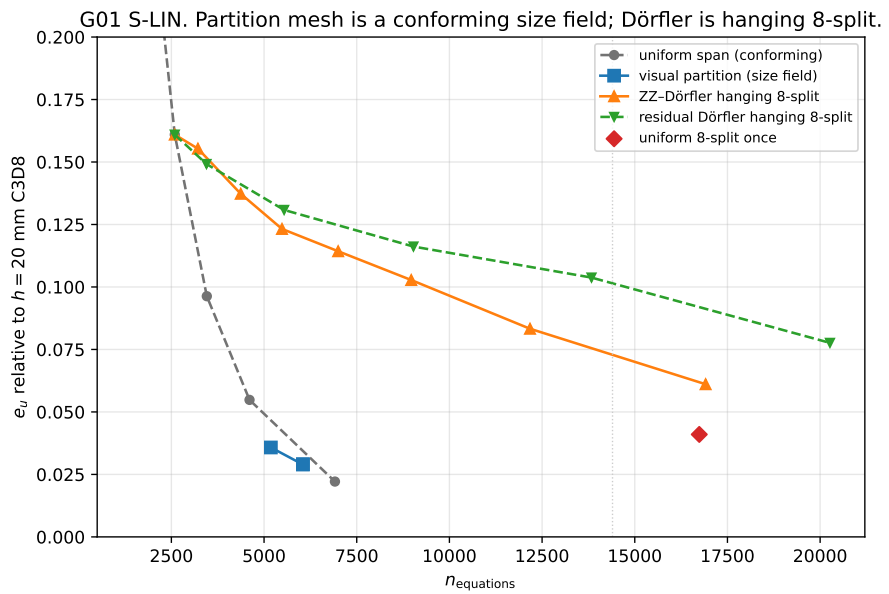


Figure 2. G01 S-LIN, e_u against $n_{\text{equations}}$. Partition points are a conforming size field; Dörfler points are hanging 8-split. Uniform spanwise ladders and one uniform 8-split are not marking methods.

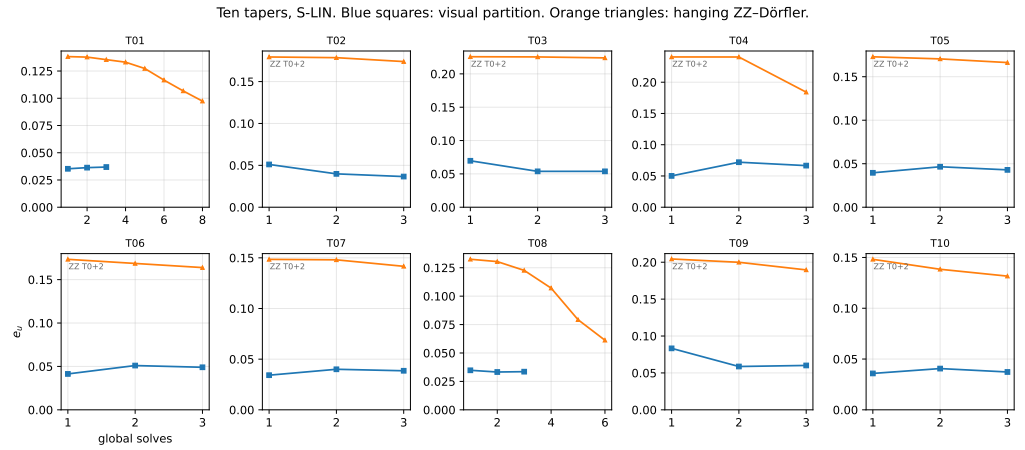


Figure 3. Ten tapers, S-LIN, global-solve axis. Blue: visual partition (three solves). Orange: hanging ZZ-Dörfler. Panels with only T0+2 Dörfler points are existence runs, not full curves.

4.5. Local prediction (R8), not a partition

On G01 T0 the four conformal seeds are eid 2 (clamp web), 49 (re-entrant), 243 (mid-span), 532 (tip flange). In-process Q2 has $n_{\text{free}} = 9, 13, 11, 9 > 0$ and finite ΔE_p . Unenriched in-process C3D8 ALLSE matches the assembler. In-process h ALLSE matches a CalculiX 8-split local job to relative error $< 10^{-4}$ on the first three seeds. The tip seed's potential comparison is dominated by the load term on Dirichlet nodes; ALLSE on that seed still drops. After one 8-split of the re-entrant hex, a fine hex (eid 19) has four hanging-face neighbours. In-process ΔE_h and ΔE_p on that hanging patch are finite. The Houston–Wihler outer loop to nine cycles is not completed in this artefact set; R8 is the gate that was run. Local prediction is not drawn as a partition in any figure.

4.6. Nonlinear cards

The twelve required nonlinear cells (G01, T01, T08 $\times$ S-ISO, S-KIN, D-LIN, D-NLG) all produced three CalculiX jobs. Distinct N holds on G01 S-ISO (5040, 5616, 5904), T01 S-ISO, T01 D-LIN, T08 S-ISO, T08 D-LIN and T08 D-NLG. The others collide on at least two of the three compiles. Tip (or peak) displacement moves with N but is not ranked: this campaign has no nonlinear fine reference, so e_u is not defined on these cards. All twelve cells remain solvable.

5. Discussion

A previous implementation bisected every spanwise station that contained a marked hex. Two such cuts stacked degrees of freedom at the clamp, and tip displacement—dominated by root rotation—looked converged. That is not Dörfler 1996. With hanging 8-splits the G01 ZZ curve is still at 13.7% after three global solves and at 6.1% when N crosses the reference budget. The visual partition, which places size at the clamp, the re-entrant corner and the tip flange on the first rebuild, sits at a few percent after one to three solves on the same error axis.

That does not say the partition is a cheaper complete AFEM oracle. On the N axis a uniform ladder at $N = 6912$ already has $e_u = 2.21\%$, below the collided PSO point at $N = 6048$. The partition's claim, on this campaign, is about sample complexity of global solves and about putting size on named structural parts before a bulk marker has walked out of the clamp. Several tapers show a revision step that does not reduce e_u (T01, T04, T05). $\kappa \approx 0$ on several cards is consistent with an embedded family after the regional step has already moved sizes; it is reported, not treated as a failure by itself. PSO collisions (rev1 and PSO compiling the same N) are failures of the three-point gate and are not redrawn.

The six-zone spanwise template (X0–X5, 0–120 ... 1700–2000 mm) is a repository artefact dated 2026-08-15. It is not Houston–Wihler, not Dörfler, and not the visual partition of this paper. It is not used in the experiments, not given a section, and not given an appendix.

6. Conclusions

LLM visual partitioning and per-element marking were run on the same solid T-beam family, with Dörfler implemented as hanging 8-splits of marked hexes and Houston–Wihler implemented as per-element local Dirichlet problems. On G01 S-LIN, three partition solves reach $e_u \approx 2.9\%$ while three Dörfler solves remain above 13%. The Dörfler curve was continued to eight points. Local prediction is a third, more expensive per-element method; R8 shows a solvable Q2 p -branch and a hanging-aware patch. No six-zone method is claimed. No Euler–Bernoulli error axis is used. T01 is not G01.

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1. Zienkiewicz, O.C.; Zhu, J.Z. A simple error estimator and adaptive procedure for practical engineering analysis. *Int. J. Numer. Methods Eng.* **1987**, *24*, 337–357. <https://doi.org/10.1002/nme.1620240206>.
2. Dörfler, W. A convergent adaptive algorithm for Poisson’s equation. *SIAM J. Numer. Anal.* **1996**, *33*, 1106–1124. <https://doi.org/10.1137/0733054>.
3. Chamoin, L.; Legoll, F. An introductory review on a posteriori error estimation in finite element computations. *SIAM Rev.* **2023**, *65*, 963–1028.
4. Houston, P.; Wihler, T.P. Adaptive energy minimisation for hp -finite element methods. *Comput. Math. Appl.* **2016**, *71*, 977–990. <https://doi.org/10.1016/j.camwa.2016.01.002>.
5. Morin, P.; Nochetto, R.H.; Siebert, K.G. Data oscillation and convergence of adaptive FEM. *SIAM J. Numer. Anal.* **2000**, *38*, 466–488.
6. Cascón, J.M.; Kreuzer, C.; Nochetto, R.H.; Siebert, K.G. Quasi-optimal convergence rate for an adaptive finite element method. *SIAM J. Numer. Anal.* **2008**, *46*, 2524–2550.
7. Amor-Martin, A.; Garcia-Castillo, L.E. Adaptive semi-structured mesh refinement techniques for the finite element method. *Appl. Sci.* **2021**, *11*, 3683. <https://doi.org/10.3390/app11083683>.
8. Babuška, I.; Rheinboldt, W.C. Error estimates for adaptive finite element computations. *SIAM J. Numer. Anal.* **1978**, *15*, 736–754. <https://doi.org/10.1137/0715049>.
9. Bammer, P.; Schröder, A.; Wihler, T.P. An hp -adaptive strategy based on locally predicted error reductions. *Comput. Methods Appl. Math.* **2025**, *25*, 777–805. <https://doi.org/10.1515/cmam-2023-0250>.
10. Yang, J.; Mittal, S.; Schaub, T. Reinforcement learning for adaptive mesh refinement. In Proceedings of the Proc. AISTATS, 2023.
11. Foucart, C.; Charous, A.; Lermusiaux, P.F.J. Deep reinforcement learning for adaptive mesh refinement. *J. Comput. Phys.* **2023**, *491*, 112381.